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Subject Code	ISYS2800
Subject Name	Business Analytics
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Assessment number &title	**Final Assessment** Individual Reflective Writing: Applying Data Analytics Responsibly

I declare that all material in this assignment/project/tutorial is my own work except where there is clear acknowledgement or reference to the work of others, and I have complied with and agreed to the University statement on Plagiarism and Academic Integrity.

Signature: Mohamed ……

Date: 28/9/2026 ……

Assessor's feedback:	
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Total Marks:	

Introduction

I have learned how to apply practical, applied computer skills to make business decisions. Excel, Power BI, and Tableau allowed me to organize and effectively present data; the house price regression assignment made supervised machine learning easier to understand. Perhaps the greatest value of this course was realizing that generating an output (i.e., answering) and providing a justification for a decision (i.e., explaining why you answered that way) are separate responsibilities. This self-assessment will be conducted using three sources: my comparison of Excel and Power BI in CW2, the RavenStack group project and the Starbucks Case Study from CW3, as well as class material. These areas of study can serve as a base for determining where I am technically, what limits exist on analytic techniques, and where I still need to develop professionally.

Development of data analysis skills

One of the most valuable lessons from comparing Excel and Power BI in CW2 is that data preparation needs to be explained regardless of whether there are any errors (duplicates) found in your data. Conducting a duplicate check simply tells you one thing: that there were no duplicate entries, but nothing else. Instead of saying the entire data set is clean because I did not find any duplicates, I could have documented how much of the data set I actually cleaned.

The comparison between these two tools helped me think about selecting the right tool based on what I am doing with the data. I liked how Excel allowed me to see and manipulate single values in formulas, how Power BI provided links between different visual elements, and how it performed repetitive steps in preparing my data. However, when I previously said that Excel's manual cleaning of data was better than Power BI's automation of data cleaning, I oversimplified things. Power Query is available in both Excel and Power BI and allows you to save your transformations so they can be refreshed automatically (Microsoft, 2026). Therefore, I should select which workflow to use by thinking about the type of reporting needed by the user(s), if the workflow needs to be repeated over time, and whether one product provides a superior method for cleaning/preparing data than the other product.

In the RavenStack Group Project, my contributions to our team were in creating our datasets through preparation and cleaning, establishing relationships among the models, and assisting in adding visuals to our dashboards and writing DAX measures. Since we combined account,

subscription, usage, support, and churn information into one unit of analysis, account ID became very important. Because multiple activities can occur within one account, counting joined rows can provide incorrect numbers for total customers. In future projects, I will compare the number of unique accounts prior to modeling versus after modeling, identify any unmatched identifiers, and test all filter selections to ensure they are producing the correct numbers. As such, data modeling becomes a series of explanations/justifications as opposed to a simple graphical representation of related tables.

Application of analytical methods and models

The house price exercise helped me understand why multiple linear regression belongs to supervised learning. House size, bedrooms, bathrooms, age, distance from the city centre, and parking spaces are predictors; selling price in RM is the numerical target. The model estimates a price from their combined relationships. Ordinary least squares fits coefficients by minimising squared differences between observed and predicted values (scikit-learn developers, n.d.-b). A property business could use this as an initial pricing aid, but a prediction should support a valuation discussion rather than determine a selling price automatically.

My next step is to distinguish between being able to explain the model and proving it works well. Count of bedrooms & square footage may be interdependent in regard to the information they provide. Therefore, individual coefficients will likely be unstable. Additionally, relationships may vary by location or price range. I would evaluate predictions on unseen properties, compare them with a simple baseline, & report mean absolute error in RM. Training & testing on the same observations can give misleadingly optimistic results. (scikit-learn developers, n.d.-a)

The limitations illustrated by the RavenStack analysis are somewhat different. According to the RavenStack analysis, there were 110 of 500 account churning events. The RavenStack Group utilized a rules-based score to determine which of the remaining 78 active accounts to contact first. While a score of 100 does not represent a 100% chance of churning, that number will at least allow you to organize follow-up activity with those contacts. Likewise, while the higher rate of churning associated with one specific industry indicates a trend to investigate, the data itself does not explain what drives that trend.

In addition, CW3 further cautioned regarding the Starbucks offer analysis. In particular, the analysis distinguished between completions that occurred immediately after viewing an offer

versus completions that did not occur until later. Although this is much better than simply treating all completions identically, using "influenced" to label completions after viewing an offer can be misleading since a customer may have viewed an offer and purchased regardless. Therefore, I would suggest using "completed after viewing an offer" when communicating with stakeholders about the results of this study and conducting a controlled test prior to claiming additional revenue as a result of these offers.

Finally, my experience in reviewing the master's student's presentation on Naive Bayesian Detection of Pig Butchering Scams provided me another contrast -- classification instead of predicting numerical values. Naive Bayes assumes that each feature is independent of the other features given the class (scikit-learn developers, n.d.-c). Like many helpful methodologies, Naive Bayes relies upon some assumptions.

Communication of findings and recommendations

CW2 demonstrates the necessity of separating a discovery from an explanation. A large number of furniture sales is certainly no guarantee of high profits. However, the conclusion that the reason for this was the overuse of price cuts (discounts) has yet to be supported with evidence regarding discount use, cost, or product mix. It is now my suggestion to examine these areas before establishing discount use as a causal factor. Monthly peaks during the previous year may provide some ideas for how to plan ahead but they are too little information on their own to confirm that the same type of peak will occur in future years.

The Starbucks Tableau dashboard is an excellent representation of how to communicate effectively. Using linked views, a stakeholder will be able to drill down into a specific segment; however, multiple charts can compete for the viewers' attention. In terms of design, Tableau (n.d.) states that you should build your dashboards based on who will use them and limit the number of views.

For a loyalty manager, I would present my decision on what offer I think should be tested and then display the appropriate comparative chart and what its limitations are. Finance will also want information related to the rewards cost and margins.

A comment regarding the value of the rewards being released prior to having a documented view does not equate to a promise of recoverable profit.

Clear titles, labelled filters, and a visible disclaimer stating that the data used in this simulation has been generated artificially will help provide a more accurate recommendation and allow others to question the recommendation with greater ease.

Ethical and professional learning

The Starbucks case gives us a tangible definition of what it means to be fair. Just because you're not using demographics in your cluster doesn't mean the results will be fair; behavior can simply represent income (or lack thereof) or opportunity for participation. Thus, the report's disparate income profiles present an audit concern and do not provide proof of discrimination on their own. Prior to making decisions about who gets targeted offers, I would first review which applicants receive benefits and who are denied based on an unexpected difference in those groups, and secondly refrain from assuming missing demographic information automatically disqualifies eligibility.

The scam detection presentation demonstrates responsibility in addition to just being accurate. The same false accusation could ruin the reputation of a genuine user, whereas missing a scam could put a person at risk. If there was going to be any evaluation of how well I would perform the task, it would have been done in conjunction with human oversight, and all communication regarding me personally would have had to remain confidential; access would have only existed as necessary for completing the task. Those were my thoughts on the application's potential and should not be confused with comments about how the presenter implemented his/her application. Tabassi’s (2023) risk management framework supports considering potential negative impacts when using an artificial intelligence system from beginning to end. That will mean minimizing identifiable information within my own work, documenting any limitations, and identifying who would be accountable for taking action based on what the AI produces.

Challenges and strategies for improvement

Practical training with Excel, Power BI, and Tableau was useful for my learning; however, replicating a series of steps does not necessarily equate to an explanation of how those steps work. I found that the house prices project provided very practical examples where the input variables and output (target) are both relevant to a common issue. This also provides me with a method of testing if I understand something by providing a description of the intended use of

each input variable, describing what a given prediction means, and determining when a prediction may be misleading.

In reviewing my coursework, I have found an area where there is room for development in my analytical writing: when making recommendations based on patterns, I sometimes move too rapidly. Two examples of this were with the "discount" explanation in CW2 and the "influenced" label in CW3. My plan for developing this skill will be to develop three distinct statements for every finding I make: (1) What did you see?, (2) What could cause this?, and (3) What do I need to see to take action? I will practice rewriting a small analysis by myself, as well as maintaining records of the decisions made during the process and checking total values against a basic self-verification calculation. There may be holes within a polished dashboard that my rapid progress through the course has concealed.

Future application of business analytics

I would also alter the way I obtain the data. Prior to importing a new data file, I would take note of the data file's origin (where it came from), date range (coverage period), and whether any permission is required in order to use this data, as well as what each row represents. Although the public and simulated datasets provided by the coursework were convenient for me to gain experience with, they do not provide any indication that either one represents a specific company. In an actual project, I would collaborate with the owner of the data and reach agreement about any missing data and what decision the analysis will be attempting to assist with prior to selecting which charts/models I will utilize.

I'd like to utilize my skills for business reporting and decision-making by improving my ability to evaluate models over the next few months. In the first month, I will recreate a small retail report that includes documented clean checks of data and total reconciliation. In the second month, I will run the house prices again on a different test set and calculate the error using RM. After this, I will practice communicating one recommendation to a person without technical knowledge, and have the person communicate back about what they believe is the recommended course of action and how uncertain it is. To me, success would be producing a reproducible analysis, with a recommendation the reader can understand as opposed to simply creating yet another dashboard. As for a practical business application, I will start with a small pilot, review results, and adjust as needed based on new evidence.

Conclusion

ISYS2800 has given me practical examples of gathering, preparing, analysing, and communicating data. My coursework and classroom activities also provide reasons to question apparently convincing outputs: clean-looking records may need more checks, model scores may be misunderstood, and observed behaviour may not establish causation. The development I value most is connecting technical work with these questions. My future practice should combine reproducible preparation, appropriate evaluation, clear communication, and attention to the people affected by decisions. These habits would make my business analytics work more useful and more responsible.

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